The table is entered by using three-word sets. W1 is the flow area or stem position and must be normalized. The factor W1 on card CCC0400 can be used to normalize the flow area or stem position. In either case, the implication is that if the valve is fully closed, the normalized term is 0. If the valve is fully open, the normalized term is 1.0. Any value may be input that is between 0.0 and 1.0. The forward and reverse flow coefficients are W2 and W3, respectively. The code internally converts flow coefficients to energy loss coefficients by the formula $K = 2 \bullet A_j^2/(\rho \bullet CSUBV^2)$, where ρ is a reference density of 991.091 kg/m^3 (62.3706 lb$_m$/ft^3), A_j is the full open valve area, and CSUBV is the flow coefficient. On card CCC0400, W2 may be used to modify the definition of CSUBV. A smooth area change must be specified in W6 on card CCC0101 to use the CSUBV table. CSUBV is entered in British units only.

W1(R) Normalized flow area or normalized stem position (See Section 8.15.10 for motor valve and Section 8.15.11 for servo valve).

W2(R) Forward CSUBV $\{(gal/min)/[(lb_f/in^2)^{0.5}]\}$. The CSUBV is input in British units only and is converted automatically by the code to SI units [(m$^{7/}$kg)$^{0.5}$] using a multiplier of 7.598055E-7 as the conversion factor.

W3(R) Reverse CSUBV $\{(gal/min)/[(lb_f/in^2)^{0.5}]\}$.

The user is cautioned that when CSUBV = 0.0, the valve acts like a check valve and no flow occurs through the valve even if the normalized stem position is 1.0 or open.

8.16 Pump Component, PUMP

A pump component is indicated by PUMP for Word 2 on card CCC0000. A pump consists of one volume and two junctions, one attached to each end of the volume. For major edits, minor edits, and plot variables, the volume in the pump component is numbered as CCC010000. The pump junctions are numbered CCC010000 for the inlet junction and CCC020000 for the outlet junction.

8.16.1 Pump Volume Geometry, CCC0101-0107

This card (or cards) is required for a pump component. The seven words can be entered on one or more cards, and the card numbers need not be consecutive.

W1(R) Area (m^2, ft^2).

W2(R) Length (m, ft).

W3(R) Volume (m^3, ft^3). The program requires that the volume equals the area times the length (W3 = W1•W2). At least two of the three quantities, W1, W2, W3, must be nonzero. If one of the quantities is 0, it will be computed from the other two. If none of the words are 0, the volume must equal the area times the length within a relative error of 0.000001.

W4(R) Azimuthal angle (degrees). The absolute value of this angle must be $\leq$ 360 degrees. This quantity is not used in the calculation but is specified for possible automated drawing of nodalization diagrams.

W5(R) Inclination angle (degrees). The absolute value of this angle must be $\leq$ 90 degrees. The angle 0 degrees is horizontal, and positive angles have an upward direction, i.e., the outlet is at a higher elevation than the inlet. This angle is used in the interphase drag calculation.

W6(R) Elevation change (m, ft). A positive value is an increase in elevation. The absolute value of this quantity must be equal to or less than the length. If the vertical angle orientation is 0, this quantity must be 0. If the vertical angle is nonzero, this quantity must also be nonzero and have the same sign. For this component, this Word 6 is not compared to the length (Word 2) to decide if the horizontal or vertical flow regime is used. Rather, the pump flow regime map is used.

W7(I) Volume control flags. This word has the format tlpvbfe. It is not necessary to input leading zeros. Volume flags consist of scalar oriented and coordinate direction oriented flags. Only one value for a scalar oriented flag is entered per volume but up to three coordinate oriented flags can be entered for a volume, one for each coordinate direction. At present, the f flag is the only coordinate direction oriented flag. This word enters the scalar oriented flags and the x-coordinate flag. The pump component forces all volume flags except for the e digit, and y- and z-coordinate flags are not read. The effective format is 000000e.

e-flag The equilibrium digit e specifies if nonequilibrium or equilibrium is to be used; e = 0 specifies that a nonequilibrium (unequal temperature) calculation is to be used, and e = 1 specifies that an equilibrium (equal temperature) calculation is to be used. Equilibrium volumes should not be connected to nonequilibrium volumes. The equilibrium option is provided only for comparison to other codes.

8.16.2 Pump Inlet Junction, CCC0108

This card is required for a pump component.

W1(I) Volume code of connecting volume on inlet side. This refers to the component from which the junction coordinate direction originates. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N indicates the face number. The number N equal to 1 and 2 specifies the inlet and outlet faces respectively for the volume's coordinate direction. The number N equal to 3-6 specifies crossflow. The number N equal to 3 and 4 would specify inlet and outlet faces for the second coordinate direction; N equal to 5 and 6 would do the same for the third coordinate direction. For connecting to a time-dependent volume, only the number N equal to 1 or 2 is allowed.

W2(R) Area (m^2, ft^2). If 0, the area is set to the minimum area of adjacent volumes. If an abrupt area change, the area must be equal to or less than the minimum of the adjacent volume areas. If a smooth area change, no restrictions exist.

W3(R) Reynolds number independent forward flow energy loss coefficient, A_F. This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. A variable loss coefficient may be specified (see Section 8.16.6). The interpretation and use of this loss coefficient vary based upon the “a” junction control flag selected in W5. If the smooth area change option is selected (a = 0), the entry here is the loss coefficient appropriate for the junction flow area resulting from the entry made in W2. If one of the abrupt area change options is selected (a = 1 or 2), the entry here is the loss coefficient appropriate for the minimum flow area between the two hydrodynamic volumes connected by this junction. If the a = 1 option is selected, a loss coefficient entered here is considered to be additive with the abrupt area change flow energy loss coefficient, which is calculated separately by the code.

W4(R) Reynolds number independent reverse flow energy loss coefficient, A_R. This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is negative. A variable loss coefficient may be specified (see Section 8.16.6). See information for the previous word regarding interpretation and use of this loss coefficient.

W5(I) Junction control flags. This word has the format jefvcahs. It is not necessary to input leading zeros. For the PUMP inlet junction, the junction control flag is restricted to the format 00f0cah0.

f-flag The digit f specifies CCFL options. f = 0 means that the CCFL model will not be applied, and f = 1 means that the CCFL model will be applied.

c-flag The digit c specifies choking options. c = 0 means that the choking model will be applied, and c = 1 means that the choking model will not be applied.

The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske critical flow model is active. If card 1 is present and it contains Option 50, the original RELAP5 critical flow model is active.

Unlike junctions for certain other hydrodynamic components, only the default values of discharge coefficients and factors used in the critical flow models are allowed for junctions in the PUMP component. Problem renodalization should be used to circumvent this limitation should it prove to be significant.

For the Henry-Fauske critical flow model, the default value for the discharge coefficient is 1.0 and the default value for the thermal nonequilibrium constant is 0.14. For the original RELAP5 critical flow model, the default values for the subcooled, two-phase and superheated discharge coefficients each are 1.0.

a-flag The digit a specifies area change options. a = 0 means a smooth area change and only the user supplied K_{loss} is applied. a = 1 means full abrupt area change model and the code calculates forward and reverse contraction/expansion K_{loss} terms and adds them to the user supplied K_{loss} terms. In addition, the code includes extra interphase drag to account for the vena contracta. a = 2 means a partial abrupt area change model and it is the same as ¨a = 1 except no code calculated forward and reverse contraction/expansion K_{loss} is applied.

h-flag The digit h specifies nonhomogeneous or homogeneous. h = 0 specifies the nonhomogeneous (two-velocity momentum equations) option; h = 1 or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option (h = 1 or 2), the major edit printout will show a 1.

8.16.3 Pump Outlet Junction, CCC0109

This card is required for a pump component. The format for this card is identical to card CCC0108 except data are for the outlet junction.

This card is required for a pump component.

W1(I) Volume code of connecting volume on outlet side. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N indicates the face number. The number N equal to 1 and 2 specifies the inlet and outlet faces respectively for the volume's coordinate direction. The number N equal to 3-6 specifies crossflow. The number N equal to 3 and 4 would specify inlet and outlet faces for the second coordinate direction; N equal to 5 and 6 would do the same for the third coordinate direction. For connecting to a time-dependent volume, N is restricted to 1 or 2.

W2(R) Area (m^2, ft^2). If 0, the area is set to the minimum area of adjacent volumes. If an abrupt area change, the area must be equal to or less than the minimum of the adjacent volume areas. If a smooth area change, no restrictions exist.

W3(R) Reynolds number independent forward flow energy loss coefficient, A_F. This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. A variable loss coefficient may be specified (see Section 8.16.6). The interpretation and use of this loss coefficient vary based upon the “a” junction control flag selected in W5. If the smooth area change option is selected (a = 0), the entry here is the loss coefficient appropriate for the junction flow area resulting from the entry made in W2. If one of the abrupt area change options is selected (a = 1 or 2), the entry here is the loss coefficient appropriate for the minimum flow area between the two hydrodynamic volumes connected by this junction. If the a = 1 option is selected, a loss coefficient entered here is considered to be additive with the abrupt area change flow energy loss coefficient, which is calculated separately by the code.

W4(R) Reynolds number independent reverse flow energy loss coefficient, A_R. This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is negative. A variable loss coefficient may be specified (see Section 8.16.6). See information for the previous word regarding interpretation and use of this loss coefficient.

W5(I) Junction control flags. This word has the format jefvcahs. It is not necessary to input leading zeros. For the PUMP outlet junction, the junction control flag is restricted to the format 00f0cah0.

f-flag The digit f specifies CCFL options. f = 0 means that the CCFL model will not be applied, and f = 1 means that the CCFL model will be applied.

c-flag The digit c specifies choking options. c = 0 means that the choking model will be applied, and c = 1 means that the choking model will not be applied.

The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske critical flow model is active. If card 1 is present and it contains Option 50, the original RELAP5 critical flow model is active.

Unlike junctions for certain other hydrodynamic components, only the default values of discharge coefficients and factors used in the critical flow models are allowed for junctions in the PUMP component. Problem renodalization should be used to circumvent this limitation should it prove to be significant.

For the Henry-Fauske critical flow model, the default value for the discharge coefficient is 1.0 and the default value for the thermal nonequilibrium constant is 0.14. For the original RELAP5 critical flow model, the default values for the subcooled, two-phase and superheated discharge coefficients each are 1.0.

a-flag The digit a specifies area change options. a = 0 means a smooth area change and only the user supplied K_{loss} is applied. a = 1 means full abrupt area change model and the code calculates forward and reverse contraction/expansion K_{loss} terms and adds them to the user supplied K_{loss} terms. In addition, the code includes extra interphase drag to account for the vena contracta. a = 2 means a partial abrupt area change model and it is the same as

¨a = 1 except no code calculated forward and reverse contraction/expansion K_{loss} is applied.

h-flag The digit h specifies nonhomogeneous or homogeneous. h = 0 specifies the nonhomogeneous (two-velocity momentum equations) option; h = 1 or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option (h = 1 or 2), the major edit printout will show a 1.

8.16.4 Pump Inlet Junction Diameter and CCFL Data, CCC0110

This card is optional. The defaults indicated for each word are used if the card is not entered. If this card is being used to specify only the junction hydraulic diameter for the interphase drag calculation (i.e., f = 0 in Word 5 of card CCC0108), then the diameter should be entered in Word 1 and any allowable values should be entered in Words 2-4 (will not be used). If the card is being used for the CCFL model (i.e., f = 1 in Word 5 of card CCC0108), then enter all four words for the appropriate CCFL model if values different from the default values are desired.

W1(R) Inlet junction hydraulic diameter, D_j (m, ft). This is the junction hydraulic diameter used in the CCFL correlation equation and interphase drag and must be $\geq$ 0. This number should be computed from $4.0 \bullet \left(\frac{\text{area}}{\text{wetted perimeter}}\right)$. If a 0 is entered or the default is used, the junction diameter is computed from $2.0 \bullet \left(\frac{\text{area}}{\pi}\right)^{0.5}$. See Word 2 of card CCC0108 for the junction area.

W2(R) Flooding correlation form, β. If 0, the Wallis CCFL form is used. If 1.0, the Kutateladze CCFL form is used. If between 0.0 and 1.0, Bankoff weighting between the Wallis and Kutateladze CCFL forms is used. This number must be $\geq$ 0 and $\leq$ 1. The default value is 0 (Wallis form).

W3(R) Gas intercept, c. This is the gas intercept used in the CCFL correlation (when $H_f^{1/2} = 0$) and must be > 0. The default value is 1.

W4(R) Slope, m. This is the slope used in the CCFL correlation and must be > 0. The default value is 1.

8.16.5 Pump Outlet Junction Diameter and CCFL Data, CCC0111

This card is optional. The defaults indicated for each word are used if the card is not entered. If this card is being used to just specify the junction hydraulic diameter for the interphase drag calculation (i.e., f = 0 in Word 5 of card CCC0109), then the diameter should be entered in Word 1 and any allowable values should be entered in Words 2-4 (will not be used). If the card is being used for the CCFL model (i.e., f = 1 in Word 5 of card CCC0109), then enter all four words for the appropriate CCFL model if values different from the default values are desired. The format for this card is identical to card CCC0110 except that data are for the outlet junction.

W1(R) Outlet junction hydraulic diameter, D_j (m, ft). This is the junction hydraulic diameter used in the CCFL correlation equation and interphase drag and must be $\geq$ 0. This number should be computed from $4.0 \bullet \left(\frac{\text{area}}{\text{wetted perimeter}}\right)$. If a 0 is entered or the default is used, the junction diameter is computed from $2.0 \bullet \left(\frac{\text{area}}{\pi}\right)^{0.5}$. See Word 2 of card CCC0108 for the junction area.

W2(R) Flooding correlation form, β. If 0, the Wallis CCFL form is used. If 1.0, the Kutateladze CCFL form is used. If between 0.0 and 1.0, Bankoff weighting between the Wallis and Kutateladze CCFL forms is used. This number must be $\geq$ 0 and $\leq$ 1. The default value is 0 (Wallis form).

W3(R) Gas intercept, c. This is the gas intercept used in the CCFL correlation (when $H_f^{1/2} = 0$) and must be > 0. The default value is 1.

W4(R) Slope, m. This is the slope used in the CCFL correlation and must be > 0. The default value is 1.

8.16.6 Pump Inlet Junction Form Loss Data, CCC0112

This card is optional. The user-specified form loss is given in Words 3 and 4 of card CCC0108 if this card is not entered. If this card is entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F Re^{-C_F}$$

$$K_R = A_R + B_R Re^{-C_R}$$

where K_F and K_R are the forward and reverse form loss coefficient. A_F and A_R are the Words 3 and 4 of card CCC0108. Re is the Reynolds number based on mixture fluid properties. If this card is being used for the form loss calculation, then enter all four words for the appropriate expression. See the discussion regarding the flow area basis assumed for A_F and A_R; the same information applies for B_F and B_R.

W1(R) B_F ($\geq$ 0). This quantity must be greater than or equal to 0.

W2(R) C_F ($\geq$ 0). This quantity must be greater than or equal to 0.

W3(R) B_R ($\geq$ 0). This quantity must be greater than or equal to 0.

W4(R) C_R ($\geq$ 0). This quantity must be greater than or equal to 0.

8.16.7 Pump Outlet Junction Form Loss Data, CCC0113

This card is optional. The user-specified form loss is given in Words 3 and 4 of card CCC0109 if this card is not entered. If this card is entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F Re^{-C_F}$$

$$K_R = A_R + B_R Re^{-C_R}$$

where K_F and K_R are the forward and reverse form loss coefficient. A_F and A_R are the Words 3 and 4 of card CCC0109. Re is the Reynolds number based on mixture fluid properties. If this card is being used for the form loss calculation, then enter all four words for the appropriate expression. See the discussion regarding the flow area basis assumed for A_F and A_R; the same information applies for B_F and B_R. The format of this card is identical to card CCC0112 except data are for the outlet junction.

W1(R) B_F ($\geq$ 0). This quantity must be greater than or equal to 0.

W2(R) C_F ($\geq$ 0). This quantity must be greater than or equal to 0.

W3(R) B_R ($\geq$ 0). This quantity must be greater than or equal to 0.

W4(R) C_R ($\geq$ 0). This quantity must be greater than or equal to 0.

8.16.8 Pump Volume Initial Conditions, CCC0200

This card is required for a pump component.

W1(I) Control word. This word has the format $\underline{\varepsilon}$bt. It is not necessary to input leading zeros.

e-flag The digit $\underline{\varepsilon}$ specifies the fluid, where $\underline{\varepsilon}$ = 0 is the default fluid. The value for $\underline{\varepsilon}$ > 0 corresponds to the position number of the fluid type indicated on the 120-129 cards (i.e., $\underline{\varepsilon}$ = 1 specifies H_2O, $\underline{\varepsilon}$ = 2 specifies D_2O, etc.). The default fluid is that set for the hydrodynamic system by cards 120-129 or this control word in another volume in this hydrodynamic system. The fluid type set on cards 120-129 or these control words must be consistent (i.e., not specify different fluids). If cards 120-129 are not entered and all control words use the default $\underline{\varepsilon}$ = 0, then H_2o is assumed to be the fluid. It is recommended that the user use the 120-129 cards to set the fluid type in the systems.

b-flag The digit b specifies whether boron is present. b = 0 specifies that the volume liquid does not contain boron; b = 1 specifies that a boron concentration in mass of boron per mass of liquid (which may be 0) is being entered after the other required thermodynamic information.

t-flag The digit t specifies how the following words are to be used to determine the initial thermodynamic state. t = 0-3 specifies one component (steam/water). Entering t = 4-6 allows the specification of two components (steam/water and noncondensable gas).

With options t equal to 4-6, names of the components of the noncondensable gas must be entered on card 110, and mass fractions of the noncondensable gas components are entered on card 115.

One Component (Steam/Water), Equilibrium or Non-equilibrium

[P,U_f,U_g,α_g] If t = 0, the next four words are interpreted as pressure (Pa, lb_f/in^2), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), and vapor void fraction. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. Enter boron in W6 if needed.

One Component (Steam/Water), Equilibrium

[T,x_s] If t = 1, the next two words are interpreted as temperature (K, ^oF) and static quality in equilibrium condition. Enter boron in W4 if needed.

[P,x_s] If t = 2, the next two words are interpreted as pressure (Pa, lb_f/in^2) and static quality in equilibrium condition. Enter boron in W4 if needed.

[P,T] If t = 3, the next two words are interpreted as pressure (Pa, lb_f/in^2) and temperature (K, ^oF) in equilibrium condition. Enter boron in W4 if needed.

Two Components (Steam/Water/Air), Non-equilibrium

The following options are used for input of non-equilibrium two-phase and/or noncondensable states. In all cases, the criteria used for determining the range of values for static quality (x_s) are

Two-phase if $1.0E\text{-}9 \leq x_s \leq 0.99999999$,

Single phase if $x_s < 1.0\ E\text{-}9$ or $x_s > 0.99999999$

The static quality is given by $x_s = M_g/(M_g + M_f)$, where $M_g = M_s + M_n$.

[P,T,x_s] If t = 4, the next three words are interpreted as pressure (Pa, $lb_f/in.^2$), temperature (K, ^oF), and static quality in equilibrium condition. Using this input option with static quality > 0.0 and $\leq$ 1.0, saturated noncondensables will result. Also, the temperature is restricted to be less than the saturation temperature at the input pressure. Setting static quality to 0.0 is used as a flag that will initialize the volume to all noncondensables (dry noncondensable) with no temperature restrictions. Static quality is reset to 1.0 using this dry noncondensable option. Enter boron in W5 if needed.

[T,x_s,x_n] If t = 5, the next three words are interpreted as steam saturation temperature (K, ^oF), static quality, and noncondensable quality in equilibrium condition. Enter boron in W5 if needed. Both the static and noncondensable qualities are restricted to be between 1.0 E-9 and 0.99999999. The liquid will be subcooled. The input temperature determines the steam partial pressure, and the total pressure may be greater than the critical pressure, which will cause an input processing error. Little experience has been obtained using this option, and it has not been checked out.

[P,U_f,U_g,α_g,x_n] If t = 6, the next five words are interpreted as pressure (Pa, $lb_f/in.^2$), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), vapor void fraction, and noncondensable quality. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. The combinations of vapor void fraction and noncondensable quality must be thermodynamically consistent. If noncondensable quality is set to 0.0, noncondensables are not present and the input processing branches to that type of processing (t = 0). If noncondensables are present (noncondensable quality greater than 0.0), then the vapor void fraction must also be greater than 0.0. If the noncondensable quality is set to 1.0 (pure noncondensable), then vapor void fraction must also be 1.0. When both the vapor void fraction and the noncondensable quality are set to 1.0, the volume temperature is calculated from the noncondensable energy equation using the input vapor specific internal energy. Enter boron in W7 if needed.

W2-W7(R) Quantities as described under word 1. Depending on the control word, 2-6 quantities may be required. Enter only the minimum number required. If entered, boron concentration (mass of boron per mass of liquid) follows the last required word for thermodynamic conditions.

8.16.9 Pump Inlet Junction Initial Conditions, CCC0201

This card is required for a pump component.

W1(I) Control word. If 0, the next two words are velocities; if 1, the next two words are mass flow rates.

W2(R) Initial liquid velocity or mass flow. This quantity is either velocity (m/s, ft/s) or mass flow (kg/s, lb/s).

W3(R) Initial vapor velocity or mass flow. This quantity is either velocity (m/s, ft/s) or mass flow (kg/s, lb/s).

W4(R) Initial interface velocity (m/s, ft/s). Enter 0.

8.16.10 Pump Outlet Junction Initial Conditions, CCC0202

This card is required for a pump component. This card is similar to card CCC0201 except that data are for the outlet junction.

W1(I) Control word. If 0, the next two words are velocities; if 1, the next two words are mass flow rates.

W2(R) Initial liquid velocity or mass flow. This quantity is either velocity (m/s, ft/s) or mass flow (kg/s, lb/s).

W3(R) Initial vapor velocity or mass flow. This quantity is either velocity (m/s, ft/s) or mass flow (kg/s, lb/s).

W4(R) Initial interface velocity (m/s, ft/s). Enter 0.

8.16.11 Pump Index and Option, CCC0301

This card is required for a pump component.

W1(I) Pump table data indicator. If 0, single-phase homologous tables are entered with this component. A positive nonzero number indicates that the single-phase tables are to be obtained from the pump component with this number. If -1, use built-in data for the Bingham pump. If -2, use built-in data for the Westinghouse pump. If -3, use built-in data for the Wolsong pump.

W2(I) Two-phase index. Enter -1 if the two-phase option is not to be used. Enter 0 if the two-phase option is desired and two-phase multiplier tables are entered with this component. Enter nonzero if the two-phase option is desired and the two-phase multiplier table data are to be obtained from the pump component with the number entered. There are no built-in data for the two-phase multiplier table.

W3(I) Two-phase difference table index. Enter -3 if the two-phase difference table is not needed (i.e., if W2 is -1). Enter 0 if a table is entered with this component. Enter a positive nonzero number if the table is to be obtained from pump component with this number. Enter -1 for built-in data.

W4(I) Pump motor torque table index. If -1, no table is used. If 0, a table is entered for this component. If nonzero, use the table from the component with this number.

W5(I) Time-dependent pump velocity index. If -1, no time-dependent pump rotational velocity table is used and the pump velocity is always determined by the torque-inertia equation. If 0, a table is entered with this component. If nonzero, the table from the pump component with this number is used. A pump velocity table cannot be used when the pump is connected to a shaft control component.

W6(I) Pump trip number. When the trip is off, electrical power is supplied to the pump motor; when the trip is on, electrical power is disconnected from the pump motor. The pump velocity depends on the pump velocity table and associated trip, the pump motor torque data, and this trip. If the pump velocity table is being used, the pump velocity is always computed from that table. If the pump velocity table is not being used, the pump velocity depends on the pump motor torque data and this trip. If the trip is off and no pump motor torque data are present, the pump velocity is the same as for the previous time step. This will be the initial pump velocity if the pump trip has never been set. Usually the pump trip is a latched trip, but that is not necessary. If the trip is off and a pump motor torque table is present, the pump velocity is given by the torque-inertia equation where the net torque is given by the pump motor torque data and the homologous torque data. If the trip is on, the torque-inertia equation is used and the pump motor torque is set to 0. If the pump trip number is 0, no trip is tested and the pump trip is assumed to always be off.

W7(I) Reverse indicator. If 0, no reverse is allowed; if 1, reverse is allowed.

8.16.12 Pump Description, CCC0302-0304

This card (or cards) is required for a pump component.

W1(R) Rated pump velocity (rad/s, rev/min).

W2(R) Ratio of initial pump velocity to rated pump velocity. Used for calculating initial pump velocity.

W3(R) Rated flow (m^3/s, gal/min).

W4(R) Rated head (m, ft).

W5(R) Rated torque (N·m, lb_f-ft).

W6(R) Moment of inertia (kg·m^2, lb·ft^2). This includes all direct coupled rotating components, including the master for a motor driven pump.

W7(R) Rated density (kg/m^3, lb/ft^3). If 0, initial density is used. This is the density used to generate homologous data.

W8(R) Rated pump motor torque (N·m, lb_f-ft). If this word is 0, the rated pump motor torque is computed from the initial pump velocity and the pump torque that is computed from the initial pump velocity, initial volume conditions, and the homologous curves. This quantity must be nonzero if the relative pump motor torque table is entered.

W9(R) TF2 friction torque coefficient (N·m, lb_f-ft). This parameter multiplies the speed ratio (absolute pump speed/rated speed) to the second power. The friction torque factors are summed together.

W10(R) TF0, friction torque coefficient (N·m, lb_f-ft). This is constant frictional torque.

W11(R) TF1, friction torque coefficient (N·m, lb_f-ft). This multiplies the speed ratio to the first power.

W12(R) TF3, friction torque coefficient. (N·m, lb_f-ft). This multiplies the speed ratio to the third power.

8.16.13 Pump Variable Inertia, CCC0308

This card is optional, and if this card is not entered, the pump inertia is given by Word 6 of card CCC0302 . If this card is entered, pump inertia is computed from

$$I = I_3S^3 + I_2S^2 + I_1S^1 + I_0$$

where S is the relative pump speed defined as the absolute value of the pump rotational velocity divided by the rated rotational velocity.

W1(R) Relative speed at which to use the cubic expression for inertia. When the relative speed is less than this quantity, the inertia from Word 6 of card CCC0302 is used.

W2-W5(R) I_3, I_2, I_1, I_0 (kg/m ^{2}, lb·ft ^{2}).

8.16.14 Pump-Shaft Connection, CCC0309

This card is optional, and if this card is entered, the pump is connected to a SHAFT component. The pump may still be driven by a pump motor that can be described in this component, by a turbine also connected to the SHAFT component, or from torque computed by the control system and applied to the SHAFT component. The pump speed table may not be entered if this card is entered.

W1(I) Control component number of the shaft component.

W2(I) Pump disconnect trip. If this quantity is omitted or 0, the pump is always connected to the SHAFT. If nonzero, the pump is connected to the shaft when the trip is false and disconnected when the trip is true.

8.16.15 Pump Stop Data, CCC0310

If this card is optional, and if this card is not entered, the pump will not be stopped by the program.

W1(R) Elapsed problem time for pump stop (s).

W2(R) Maximum forward velocity for pump stop (rad/s, rev/min).

W3(R) Maximum reverse velocity for pump stop (rad/s, rev/min). Reverse velocity is a negative number.

8.16.16 Pump Single-Phase Homologous Curves, CCCXX00-XX99

These cards are required only if W1 of card CCC0301 is 0. There are sixteen possible sets of homologous curve data to completely describe the single-phase pump operation, that is, a curve for each head and torque for each of the eight possible curve types or regimes of operation. Entering all sixteen curves is not necessary, but an error will occur from an attempt to reference one that has not been entered.

card numbering is CCC1100-1199 for the first curve, CCC1200-1299 for the second curve, and CCC2600-2699 for the sixteenth curve. Data for each individual curve are input on up to 99 cards, which need not be numbered consecutively.

W1(I) Curve type. Enter 1 for a head curve; enter 2 for a torque curve.

W2(I) Curve regime. The possible integer numbers and the corresponding homologous curve octants are: 1 (HAN or BAN), 2 (HVN or BVN), 3 (HAD or BAD), 4 (HVD or BVD), 5 (HAT or BAT), 6 (HVT or BVT), 7 (HAR or BAR), and 8 (HVR or BVR).

W3(R) Independent variable. Values for each curve range from -1.0 to 0.0 or from 0.0 to 1.0 inclusive. The variable is v/a for W2(I) = 1, 3, 5, or 7 and a/v for W2(I) = 2, 4, 6, or 8. If the tabular data does not span the entire range of the independent variable, end point values are used for data outside the table. This usually leads to incorrect pump performance data. Thus, entering data to cover the complete range is recommended.

W4(R) Dependent variable. The variable is h/a^2 or b/a^2 for W2(I) = 1, 3, 5, or 7 and h/v^2 or b/v^2 for W2(I) = 2, 4, 6, or 8.

Additional pairs as needed are entered on this or following cards, up to a limit of 100 pairs.

8.16.17 Pump Two-Phase Multiplier Tables, CCCXX00-XX99

These cards are required only if W2 of card CCC0301 is 0; XX is 30 and 31 for the pump head multiplier table and the pump torque multiplier table, respectively.

W1(I) Extrapolation indicator. This is not used, enter 0.

W2(R) Void fraction.

W3(R) Head or torque difference multiplier depending on table type.

Additional pairs of data as needed are entered on this or additional cards as needed, up to a limit of 100 pairs. Void fractions must be in increasing order.

8.16.18 Pump Two-Phase Difference Tables, CCCXX00-XX99

These cards are required only if W3 of card CCC0301 is 0. The two-phase difference tables are homologous curves entered in a similar manner to the single-phase homologous data. Card numbering is CCC4100-4199 for the first curve, CCC4200-4299 for the second curve, … CCC5600-5699 for the sixteenth curve. Data are the same as the data for the single-phase data except that the dependent variable is the difference between single-phase and fully degraded two-phase data.

8.16.19 Relative Pump Motor Torque Data, CCC6001-6099

These cards are required only if W4 of card CCC0301 is 0. If the pump velocity table is not being used and these cards are present, the torque-inertia equation is used. When the electrical power is supplied to the pump motor (the pump trip is off), the net torque is computed from the rated pump motor torque times the relative pump motor torque from this table and the torque from the homologous data. If the electrical power is disconnected from the pump (the pump trip is on), the pump motor torque is 0.

W1(R) Pump velocity (rad/s, rev/min).

W2(R) Relative pump motor torque.

Additional pairs as needed are added on this or additional cards, up to a maximum of 100 pairs.

8.16.20 Time-Dependent Pump Velocity Control, CCC6100

This card is required only if W5 of card CCC0301 is 0. The velocity table, if present, has priority in setting the pump velocity over the pump trip, the pump motor torque data, and the torque-inertia equation.

W1(I) Trip number. If the trip number is 0, the pump velocity is always computed from this table. If the trip number is nonzero, the trip determines which table is to be used. If the trip is off, the pump velocity is set from the trip, the pump motor torque data, and the torque-inertia equation as if this table had not been entered. If the trip is on, the pump velocity is computed from this table. If Word 2 is missing, the search variable in the table is time and the search argument is time minus the trip time. If this word is used, it takes precedence over the trip number used in Word 6 of the CCC0301 card.

W2(A) Alphanumeric part of variable request code. This quantity is optional. If present, this word and the next are a variable request code that specifies the search argument for the table lookup and interpolation. TIME can be selected, but the trip time is not subtracted from the advancement time.

W3(I) Numeric part of variable request code. This is assumed to be 0 if missing.

8.16.21 Time-Dependent Pump Velocity, CCC6101-6199

These cards are required only if W5 of card CCC0301 is 0.

W1(R) Search variable. Units depend on the quantity selected for the search variable.

W2(R) Pump velocity (rad/s, rev/min).

Additional pairs as needed are added on this or additional cards, up to a maximum of 100 pairs. Time values must be in increasing order.

8.17 Circulator Component, CIRCLTR

A circulator component is indicated by CIRCLTR for Word 2 on card CCC0000. A circulator consists of one volume and two junctions, one attached to each end of the volume. For major edits, minor edits, and plot variables, the volume in the circulator component is numbered as CCC010000. The circulatorcirculator junctions are numbered CCC010000 for the inlet junction and CCC020000 for the outlet junction.

8.17.1 Circulator Volume Geometry, CCC0101-0107

This card (or cards) is required for a circulator component. The seven words can be entered on one or more cards, and the card numbers need not be consecutive.